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Cloud Security with AWS IAM



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The screenshot shows the AWS IAM console interface during the 'Specify permissions' step of creating a new policy. The breadcrumb navigation is 'IAM > Policies > Create policy'. The left sidebar shows 'Step 1: Specify permissions' as the active step, with 'Step 2: Review and create' below it. The main area is titled 'Specify permissions' with a subtitle: 'Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.' Below this is the 'Policy editor' section, which has tabs for 'Visual', 'JSON', and 'Actions'. The 'JSON' tab is selected, showing a JSON policy document with line numbers 1 through 28. The document contains two statements: the first is an 'Allow' statement for 'ec2:*' actions on all resources with a condition for 'ec2:ResourceTag/Env' equal to 'development'; the second is a 'Deny' statement for 'ec2:Describe*' actions on all resources. To the right of the JSON editor is a sidebar with 'Add actions' (a search bar for services), a list of 'Included' services (EC2), and a list of 'Available' services (AI Operations, AMP, API Gateway, API Gateway V2, ARC Region switch, ARC Zonal Shift, ASC). At the bottom of the sidebar are buttons for 'Add a resource' and 'Add a condition (optional)'. The footer of the console shows 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information for Amazon Web Services, Inc.



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Introducing Today's Project!

Project overview

In this project, I will demonstrate how to create IAM users in AWS and give them certain levels of access. I'm doing this project to learn how to give the right people access to my projects and others just access to the finished product.

Tools and concepts

Services I used were IAM and EC2. Key concepts I learnt include how to add a user, add user groups, set up a policy and relate it to a specific user or user group and how to test whether a policy's permissions are working correctly on a user's account both the hard way and the easy way.

Project reflection

This project took me approximately 1 hour. The most challenging part was just navigating for the first time since it can be tough to find items in large lists and complex dashboards. It was most rewarding to see the policy work on the new user.



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Tags

What I did in this step

In this step, I will create two AWS EC2 instances because that will help increase computing power so more users can use the service at once.

Understanding tags

Tags are ways of marking an instance and everything in it so it's easy to tell what it's being used for, For example, you don't want to be experimenting on production grade projects, whereas you can try some things in a dev project and not worry as much about breaking something.

My tag configuration

The tag I've used on my EC2 instances is called Env. The value I've assigned for my instances are production and development.



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EC2 > Instances

Instances (2) info

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability zone
☐	nextwork-prod-saqib	i-0909616fb734f15e8	running	t3.micro	Initializing	View alarms +	us-east-2b
☐	nextwork-dev-saqib	i-0c760b231fe40f4fd	running	t3.micro	Initializing	View alarms +	us-east-2b

Select an instance

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IAM Policies

What I did in this step

In this step, I will create an IAM policy because I want to make sure only the correct people have access to certain instances. For example, I don't want an intern to have access to the production instance, since I don't want them testing things there.

Understanding IAM policies

IAM Policies are a set of rules that decide who can access what in a certain project.

The policy I set up

For this project, I've set up a policy using JSON

Policy effect

I've created a policy that allows the IAM user to do any action in the development instance except create tags or delete tags



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Understanding Effect, Action, and Resource

The Effect, Action, and Resource attributes of a JSON policy means: Action: A list of actions that are allowed or denied in an instance Effect: This basically tells us what actions the policy allows or denies Resource: Tells us what resources within an instance this policy relates to



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My JSON Policy

Specify permissions [info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

Visual JSON Actions

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "ec2:*",
7       "Resource": "*",
8       "Condition": {
9         "StringEquals": {
10          "ec2:ResourceTag/Env": "development"
11        }
12      }
13    },
14    {
15      "Effect": "Allow",
16      "Action": "ec2:Describe*",
17      "Resource": "*"
18    },
19    {
20      "Effect": "Deny",
21      "Action": [
22        "ec2:DeleteTags",
23        "ec2:CreateTags"
24      ],
25      "Resource": "*"
26    }
27  ]
28 }
```

[+ Add new statement](#)

Edit statement [Remove](#)

Add actions

Choose a service

Included

EC2

Available

AI Operations
AMP
API Gateway
API Gateway V2
ARC Region switch
ARC Zonal Shift
ASC

Add a resource [Add](#)

Add a condition (optional) [Add](#)

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Account Alias

What I did in this step

In this step, I will create an account alias because that will make it much simpler to log into the AWS account.

Understanding account aliases

An account alias is a more user friendly way of accessing an AWS account, Instead of having to remember the account ID to get to the right url, the user can just use the alias in the url instead.

Setting up my account alias

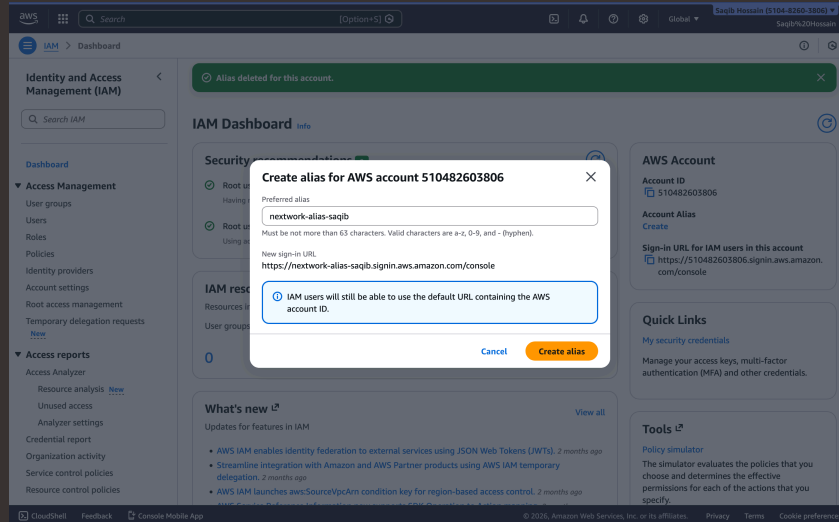
Creating an account alias took me 10 seconds. Now, my new AWS console sign-in URL is <https://nextwork-alias-saqib.signin.aws.amazon.com/console>



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IAM Users and User Groups

What I did in this step

In this step, I will set up an IAM group just for interns and an IAM user for a new intern because I want to give interns access to specific instances and not others

Understanding user groups

IAM user groups are a way of grouping users with the same permissions together so we don't have to update users individually.

Attaching policies to user groups

I attached the policy I created to this user group, which means, anyone in this user group will be only be able to do actions in the development instance.

Understanding IAM users

IAM users are users or computers within AWS that can access an account

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Logging in as an IAM User

Sharing sign-in details

The first way is by email and the 2nd is by downloading and sharing a .csv file.

Observations from the IAM user dashboard

Once I logged in as my IAM user, I noticed some of the items were showing access denied. This was because of the restrictions set up by the policy earlier.

The screenshot shows the AWS IAM console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and the user's profile (Saqib Hossain). Below the navigation bar, a green banner indicates 'User created successfully' with a 'View user' button. The main content area is titled 'Retrieve password' and shows a progress bar with four steps: 'Specify user details', 'Set permissions', 'Review and create', and 'Retrieve password' (which is the current step). The 'Retrieve password' section displays the 'Console sign-in details' for the user 'nextwork-dev-saqib'. It includes the 'Console sign-in URL' (https://nextwork-alias-saqib.signin.aws.amazon.com/console), the 'User name' (nextwork-dev-saqib), and the 'Console password' (masked with asterisks and a 'Show' button). At the bottom of the details section, there are three buttons: 'Cancel', 'Download .csv file', and 'Return to users list'.

Step 1
● Specify user details
Step 2
● Set permissions
Step 3
● Review and create
Step 4
● Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details [Email sign-in instructions](#)

Console sign-in URL
<https://nextwork-alias-saqib.signin.aws.amazon.com/console>

User name
[nextwork-dev-saqib](#)

Console password
[XXXXXXXXXXXX](#) [Show](#)

[Cancel](#) [Download .csv file](#) [Return to users list](#)



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Testing IAM Policies

What I did in this step

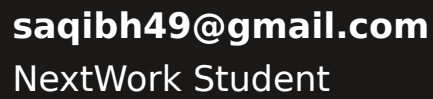
In this step, I will log into the intern's IAM user's account because I want to make sure the permissions are set up correctly to allow them to access only what I want them to access.

Testing policy actions

I tested my JSON IAM policy by attempting to make changes to my 2 instances. My policy states I can only make changes to my development instance, so when I tried to touch the production instance, it gave me an error.

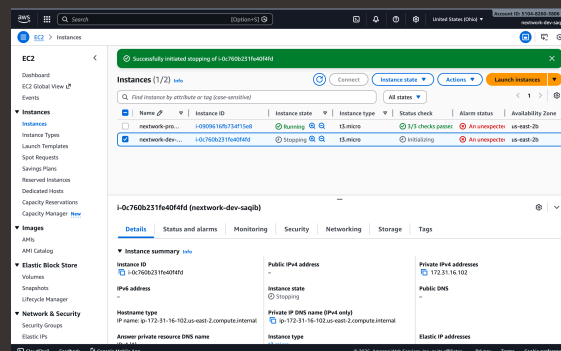
Stopping the production instance

When I tried to stop the production instance I got an error message. This was because of the permissions set in my policy from earlier.



Stopping the development instance

Next, when I tried to stop the development instance it stopped just fine because that's allowed in my policy





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IAM Policy Simulator

To extend my project, I'm going to test out my policy using a policy simulator. I'm doing this because shutting down an actual instance just to test a policy can cause problems for other users using the instance at the same time.

Understanding the IAM Policy Simulator

The IAM Policy Simulator is a safe, quick way of testing an IAM user's permissions. It's useful for making sure people have and don't have access to certain things

How I used the simulator

I set up a simulation to stop an instance and the results were that the user did not have permission. I had to adjust the specifics of the stopinstance action so it only was trying to stop instances with the tag development and it worked just fine.



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IAM Policy Simulator

Mode: Existing Policies

Saqib Hossain

Policies

Back

Editing policy: NextWorkDevEnvironmentPolicy

Customer Managed Policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "ec2:*",
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "ec2:ResourceTag/Env": "development"
        }
      }
    },
    {
      "Effect": "Allow",
      "Action": "ec2:Describe*",
      "Resource": "*"
    },
    {
      "Effect": "Deny",
      "Action": [
        "ec2:DeleteTags",
        "ec2:CreateTags"
      ],
      "Resource": "*"
    }
  ]
}
```

Apply

Reset

Policy Simulator

Amazon EC2 2 Action(s) sele... Select All Deselect All Reset Contexts Clear Results Run Simulation

Global Settings

Action Settings and Results 2 actions selected. 0 actions not simulated. 1 actions allowed. 1 actions denied.

Service	Action	Resource Type	Simulation Resource	Permission
Amazon EC2	DeleteTags	not required	*	denied 1 matching statements.
Amazon EC2	StopInstances	Instance	*	allowed 1 matching statements.



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